RadiationEffects_proj2 Slide Outline

Project overview roadmap + deep-dive lecture outline for Module 4

Planning document for building a physics-lecture-style presentation from `project_plan.pdf` and `module4_2d_ballistic_diffusive_thermal_transport.pdf`

Purpose. This document is not the final PowerPoint deck. It is a slide-planning outline for **Radiation Effects Project 2**. The first part now follows a tighter structure: **one project-overview slide plus one slide per module** (Modules 1–7). The second part remains a **10-slide deep dive into Module 4**. Each slide entry is written to help the eventual talk feel like a physics lecture rather than a project-management briefing.

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Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
BD	Ballistic-diffusive	Reduced thermal-transport model that treats boundary-originating carriers ballistically and internally scattered carriers diffusively.
BTE	Boltzmann transport equation	Microscopic transport equation for the carrier distribution function.
FEM	Finite element method	Planned numerical discretization framework for later geometric implementations.
RTA	Relaxation-time approximation	Collision model in which the distribution relaxes toward local equilibrium over a characteristic time τ.
C_j	Defect concentration	Concentration of defect species j used in Module 3 and later coupled modules.
ρ	Space-charge density	Net charge density entering Poisson's equation in Module 2.
ϕ	Electric potential	Scalar electrostatic potential solved in Module 2.
n, p	Electron and hole densities	Carrier populations used in Module 5 drift-diffusion transport.
$\boldsymbol{q}_b$	Ballistic heat flux	Boundary-memory heat-flux contribution retained in Module 4.

Symbol / acronym	Name	Meaning in this document
u_m	Medium/scattered energy	Internal-energy field associated with the scattered part of the carrier population.
Q	Volumetric heat source	Radiation-related, electrical, or externally imposed source term for Module 4.
Λ	Mean free path	Average distance traveled by a heat carrier between scattering events.
Kn	Knudsen number	Ratio $\mathrm{Kn} = \Lambda/L$, used to organize diffusive versus ballistic transport regimes.

1 How to use this slide outline

Each slide entry gives six items:

1. the main question the slide should answer,
2. the core message the audience should remember,
3. the equation or equations that should be shown,
4. a figure idea,
5. a physics drawing idea,
6. the speaker emphasis that should guide the verbal explanation.

The final slides should stay visually sparse. The derivation, physical explanation, and comparisons should mostly live in the speaker notes and in the spoken delivery.

Presentation principle. The overview part should feel like a physics-guided narrative of why each module exists, not like a software inventory. The Module 4 part should not begin with the final reduced PDE. It should begin with the physical reason Fourier conduction is insufficient, then move from microscopic heat carriers to the ballistic-diffusive split, and only then show the final 2D target model used in `RadiationEffects_proj2`.

2 Part I: Project overview slides (8 slides total)

3 Slide 1: Project overview — the full physics chain

Main question. What is the full physical story that this project is trying to model?

Core message. The project is built to explain how a radiation insult in silicon evolves into material damage, altered thermal and electrical fields, modified carrier transport, and finally degraded device behavior.

Equations to show:

$$\dot{q}_{\text{rad}}(x,y,t) \longrightarrow C_j(x,y,t) \longrightarrow \rho(x,y,t),\ \phi(x,y,t),\ T(x,y,t) \longrightarrow n, p, J_n, J_p \longrightarrow \text{device degradation.}$$

Figure idea. One end-to-end roadmap graphic with the modules laid along the physical chain from radiation deposition to device metrics.

Physics drawing idea. A silicon device cross section with an incident particle track, a localized damage region, charged defects, a thermal hot spot, bent field lines, and distorted current flow.

Speaker emphasis. Set the tone immediately: this is a mechanism-resolved physics framework. The point is not just to simulate radiation deposition, but to preserve the downstream physics that determines whether the device actually degrades.

4 Slide 2: Module 1 — radiation deposition and damage seeding

Main question. What does Module 1 provide to the rest of the framework?

Core message. Module 1 uses radiation transport to generate the initial spatial forcing: where energy is deposited, what interactions occur, and how that deposition seeds the later defect-evolution problem.

Equations to show:

$$S_{\text{dep}}(x,y,t) \equiv \text{deposited-energy or source map,}$$
$$\Phi \longrightarrow S_{\text{dep}},\ \text{interaction tallies,}\ \text{damage precursors.}$$

Figure idea. A deposition map from a particle-transport run, with track structure or localized hotspots highlighted.

Physics drawing idea. Incident particle entering silicon, showing scattering events, energy transfer, and a map of where the material first receives the radiation insult.

Speaker emphasis. Explain that Module 1 is the interface between external radiation physics and internal material/device physics. It does not yet tell us the evolved damage state; it gives the initial spatial insult.

5 Slide 3: Module 2 — electrostatics with defect-dependent space charge

Main question. How does a damaged defect population alter the electrostatic state of the device?

Core message. Module 2 translates the current charged-defect and dopant populations into electric potential and field maps through Poisson's equation.

Equations to show:

$$\begin{aligned}\nabla \cdot (\epsilon \nabla \phi) &= -\rho, \\ \rho &= q\left(p - n + N_D^+ - N_A^- + \sum_j z_j C_j\right), \\ \boldsymbol{E} &= -\nabla \phi.\end{aligned}$$

Figure idea. A 2D potential map and electric-field-line plot over a device cross section with a localized defect-rich region.

Physics drawing idea. Show charged defects locally bending field lines and perturbing depletion structure or junction fields.

Speaker emphasis. Make clear that Module 2 does not evolve the defects; it translates the current defect state into an electrostatic consequence that later feeds transport.

6 Slide 4: Module 3 — defect evolution in space and time

Main question. How does the radiation-created defect population evolve after the initial event?

Core message. Module 3 advances defect concentrations through diffusion, reaction, clustering, dissociation, and annealing, with coefficients that can depend on temperature, fields, and local material state.

Equations to show:

$$\frac{\partial C_j}{\partial t} = \nabla \cdot (D_j \nabla C_j) + R_j(C_1, C_2, \ldots, T, E).$$

$$D_j(T) = D_{0,j} \exp\left(-\frac{E_{D,j}}{k_\mathrm{B}T}\right),$$
$$k_j(T) = k_{0,j} \exp\left(-\frac{E_{A,j}}{k_\mathrm{B}T}\right).$$

Figure idea. A time sequence of defect concentration maps spreading, reacting, and relaxing.

Physics drawing idea. Vacancies and interstitials migrating, recombining, and annealing in a silicon lattice, with arrows indicating competing timescales.

Speaker emphasis. Stress that Module 3 is the engine that converts the initial deposition pattern into a changing damage landscape. Without this step, the project would assume the material damage stays frozen.

7 Slide 5: Module 4 — ballistic-diffusive thermal transport

Main question. Why is thermal transport its own module, and what makes it different from a plain heat equation?

Core message. Module 4 models heat transport in a regime where finite carrier flight lengths and boundary memory matter, so the thermal problem is richer than ordinary Fourier conduction.

Equations to show:

$$\tau \frac{\partial^2 u_m}{\partial t^2} + \frac{\partial u_m}{\partial t} = \nabla \cdot \left(\frac{k}{C} \nabla u_m\right) - \nabla \cdot \boldsymbol{q}_b + \left(Q + \tau \frac{\partial Q}{\partial t}\right).$$

Figure idea. A 2D thermal field with ballistic boundary rays and a smoother scattered-temperature background.

Physics drawing idea. Localized hot spot, outward-moving heat carriers, partial scattering, and boundary-emitted carriers contributing nonlocal thermal memory.

Speaker emphasis. Keep this slide high-level. The goal is to show why Module 4 exists in the chain and to set up the later deep dive.

8 Slide 6: Module 5 — carrier transport and recombination under damage

Main question. Once fields, temperature, and defects are known, how is device behavior predicted?

Core message. Module 5 converts the damaged material state into an electrical-performance problem by solving carrier drift, diffusion, and recombination in the perturbed field and trap landscape.

Equations to show:

$$
\begin{aligned}
J_n &= q\mu_n nE + qD_n \nabla n, \\
J_p &= q\mu_p pE - qD_p \nabla p, \\
\frac{\partial n}{\partial t} &= \frac{1}{q}\nabla \cdot J_n + G - R, \\
\frac{\partial p}{\partial t} &= -\frac{1}{q}\nabla \cdot J_p + G - R.
\end{aligned}
$$

Figure idea. A current-density map flowing around a damaged region, with local recombination or trapping zones indicated.

Physics drawing idea. Electrons and holes drifting under a bent field while defects act as recombination centers and mobility modifiers.

Speaker emphasis. This is where the materials physics becomes a device-performance prediction. Emphasize that damage matters electrically because it modifies both the driving fields and the recombination/trapping landscape.

9 Slide 7: Module 6 — self-consistent multiphysics coupling

Main question. Why can the modules not remain independent forever?

Core message. In the final framework, defects, temperature, fields, and carrier populations all feed back on one another, so the full problem must be iterated to consistency.

Equations to show:

$$\{C_j(x,y,t),\ T(x,y,t),\ \phi(x,y,t),\ n(x,y,t),\ p(x,y,t)\} \quad \text{solved iteratively to consistency.}$$

$$
\begin{aligned}
D_j &= D_j(T), \\
R_j &= R_j(T, E, C_1, C_2, \dots), \\
J_n, J_p &\text{ depend on } T, \phi, \text{ and defect-assisted recombination.}
\end{aligned}
$$

Figure idea. A coupling loop diagram among Modules 2, 3, 4, and 5.

Physics drawing idea. Circular arrows connecting defect map, temperature map, potential map, and carrier map over the same device cross section.

Speaker emphasis. Explain that the project becomes a true degradation simulator only when these feedbacks are allowed to interact rather than being treated as one-way post-processing.

10 Slide 8: Module 7 — scaling, surrogates, and practical prediction

Main question. How does the framework move beyond one small high-fidelity simulation into useful prediction?

Core message. Module 7 is the scale-bridging layer: it uses reduced-order models, surrogate models, or statistical upscaling to extend the physically grounded simulations toward larger devices, longer times, or rarer events.

Equations to show:

$$\mathcal{M}_{\text{high fidelity}} \longrightarrow \mathcal{M}_{\text{reduced/surrogate/upscaled}} \longrightarrow \text{practical prediction with quantified uncertainty.}$$

Figure idea. A high-fidelity local simulation feeding into a reduced or surrogate model that predicts behavior across a much larger device or many-device ensemble.

Physics drawing idea. Small detailed damaged-region simulation on the left, large-area chip or system-level prediction on the right, connected by an upscaling arrow.

Speaker emphasis. The project is scientifically valuable only if it can eventually inform prediction, not just produce beautiful local field maps. Module 7 is the bridge from mechanism to practical use.

11 Part II: Module 4 deep-dive slides (10 slides)

12 Slide 9: Why ordinary Fourier heat conduction is not enough here

Main question. Why does Module 4 need more than the ordinary heat equation?

Core message. Radiation-related heating can be highly localized and transient. When carrier flight distances and system scales are comparable, a purely local constitutive law loses important physics.

Equations to show:

$$\boldsymbol{q} = -k\nabla T,$$
$$C\frac{\partial T}{\partial t} = \nabla \cdot (k\nabla T) + Q.$$

Figure idea. A localized heat pulse with two conceptual predictions: an immediate smooth Fourier response versus a finite-flight-time carrier picture.

Physics drawing idea. Draw heat carriers leaving a small hot region; some scatter quickly while others travel ballistically before depositing energy elsewhere.

Speaker emphasis. The point is not that Fourier is useless. The point is that it assumes local equilibrium and local closure from the start, which can be too crude for early-time or short-length-scale transport.

13 Slide 10: Heat carriers, mean free path, and the meaning of temperature

Main question. What is actually transporting the energy, and when is temperature a subtle quantity?

Core message. Heat is carried by microscopic excitations that move a finite mean free path before scattering. Outside local equilibrium, it is often better to think in terms of equivalent internal-energy temperature rather than strict thermodynamic temperature.

Equations to show:

$$\Lambda = v\tau,$$
$$\mathrm{Kn} = \frac{\Lambda}{L},$$
$$\Delta u = C\,\Delta T.$$

Figure idea. Large-device versus small-device scattering picture, plus a boundary layer with a directional nonequilibrium distribution.

Physics drawing idea. One inset with many randomizing collisions, one inset with nearly straight carrier flights, and a separate sketch comparing isotropic and directional carrier distributions.

Speaker emphasis. This slide sets up the organizing parameter for the module: the Knudsen number tells whether the thermal problem behaves diffusively, quasi-ballistically, or strongly ballistically.

14 Slide 11: Microscopic starting point — the Boltzmann transport equation

Main question. What microscopic transport equation motivates the ballistic-diffusive model?

Core message. The correct starting point is the carrier distribution function in phase space. Streaming moves carriers through space; scattering relaxes the distribution toward local equilibrium.

Equations to show:

$$\frac{\partial f}{\partial t} + \boldsymbol{v} \cdot \nabla_{\boldsymbol{r}} f = -\frac{f - f_0}{\tau}.$$

Figure idea. A packet of heat carriers moving through the material while collisions gradually isotropize the distribution.

Physics drawing idea. Draw a local distribution function at one point deforming from a directional shape toward an isotropic equilibrium shape over a relaxation time.

Speaker emphasis. This is the deepest physics slide in the Module 4 sequence. Make clear that the reduced model is not ad hoc; it is a computationally cheaper descendant of the BTE.

15 Slide 12: Diffusive limit and why Cattaneo is still incomplete

Main question. If Fourier fails, can Cattaneo solve the problem by itself?

Core message. Fourier conduction is the strong-scattering limit of the BTE. Cattaneo improves on Fourier by introducing finite flux relaxation, but it still treats all heat transport as one local continuum field and misses boundary-originating ballistic memory.

Equations to show:

$$\boldsymbol{q} = -k\nabla T,$$
$$\tau \frac{\partial \boldsymbol{q}}{\partial t} + \boldsymbol{q} = -k\nabla T.$$

Figure idea. A comparison between Fourier's immediate-response picture, a Cattaneo thermal-wave picture, and a ballistic-arrival picture from a boundary.

Physics drawing idea. Draw one boundary-emitted carrier ray that retains memory of where it came from; contrast that with a smooth continuum wavefront.

Speaker emphasis. Say explicitly that finite propagation speed is necessary but not sufficient. Cattaneo still lacks a separate channel for boundary-emitted carriers.

16 Slide 13: The ballistic-diffusive split

Main question. How do we keep the essential nonlocal BTE physics without solving the full phase-space problem?

Core message. Split the thermal carrier population into two parts: a ballistic piece that remembers boundaries and a medium/scattered piece that can be represented by lower-order angular moments.

Equations to show:

$$I_\omega = I_{b\omega} + I_{m\omega}.$$

Figure idea. A material domain with straight boundary-originating rays for $I_{b\omega}$ and a cloud-like scattered background for $I_{m\omega}$.

Physics drawing idea. Color the ballistic carriers differently from the scattered carriers to show that the model is physically splitting the transport channels, not just algebraically rearranging the same heat equation.

Speaker emphasis. This is the conceptual heart of the module. Once the audience accepts this split, the later reduced PDE makes physical sense.

17 Slide 14: Ballistic component by characteristics and retarded boundary time

Main question. How does the ballistic part carry nonlocal boundary memory into the interior?

Core message. A ballistic carrier observed at an interior point must have left the boundary earlier and survived scattering along its path. The interior ballistic state therefore depends on retarded boundary information.

Equations to show:

$$I_{b\omega}(t, \boldsymbol{r}, \hat{\Omega}) = I_{w\omega}\left(t - \frac{s - s_0}{v}, \boldsymbol{r} - (s - s_0)\hat{\Omega}\right) \exp\left[-\int_{s_0}^{s} \frac{ds'}{v\tau_\omega}\right].$$

Figure idea. A boundary-to-interior ray with travel distance, arrival time, and exponential attenuation annotated.

Physics drawing idea. Draw several rays at different angles reaching one interior point; some arrive in time, some do not, which visually explains finite arrival-time causality.

Speaker emphasis. Emphasize that this is where the model keeps the *memory* that Fourier and Cattaneo wash out. The interior field depends on what the boundary did earlier, not just on local gradients now.

18 Slide 15: Medium/scattered component and P1 closure

Main question. Once the ballistic part is separated out, how is the scattered part simplified enough to solve?

Core message. The medium/scattered population is more isotropic than the ballistic part. A first-order angular expansion keeps the dominant isotropic term and the first directional correction.

Equations to show:

$$I_{m\omega} = J_{0\omega} + \boldsymbol{J}_{1\omega} \cdot \hat{\Omega}.$$

Figure idea. Angular distribution plot showing a nearly isotropic scattered population compared with the strongly directional ballistic population.

Physics drawing idea. Use a polar sketch: ballistic energy concentrated into narrow angular lobes, scattered energy distributed almost uniformly with a weak directional bias.

Speaker emphasis. This is the step that makes the model numerically executable. The full angular dependence is not solved directly; it is reduced to spatial fields through a controlled closure.

19 Slide 16: Final reduced ballistic-diffusive PDE

Main question. What is the actual reduced equation that Module 4 wants to solve?

Core message. The medium/scattered energy satisfies a Cattaneo-like PDE, but it is driven by the divergence of the ballistic heat flux. That forcing term is the key physics missing from ordinary local models.

Equations to show:

$$\tau \frac{\partial^2 u_m}{\partial t^2} + \frac{\partial u_m}{\partial t} = \nabla \cdot \left(\frac{k}{C} \nabla u_m \right) - \nabla \cdot \boldsymbol{q}_b + \left(\dot{q}_e + \tau \frac{\partial \dot{q}_e}{\partial t} \right).$$

Figure idea. A block diagram showing $\boldsymbol{q}_b$ computed from ballistic boundary transport, then fed into a PDE solve for u_m, with total energy assembled from ballistic and medium pieces.

Physics drawing idea. One panel for ballistic rays, one panel for the smoother medium field, and one panel showing the total temperature or energy field as the sum of both contributions.

Speaker emphasis. Do not rush this slide. Go term by term: relaxation term, medium diffusion term, ballistic forcing term, and volumetric source term.

20 Slide 17: Revised Module 4 target for `RadiationEffects_proj2`

Main question. How is the original ballistic-diffusive derivation adapted into the new 2D Module 4 for this project?

Core message. The project does not directly solve the full angular 3D problem. It uses a 2D reduced ballistic-diffusive thermal model in silicon, with volumetric radiation-related source terms and a reduced ballistic closure appropriate for a rectangular computational domain.

Equations to show:

$$\tau \frac{\partial^2 T_m}{\partial t^2} + \frac{\partial T_m}{\partial t} = \nabla \cdot (\alpha \nabla T_m) - \frac{1}{\rho_m c_p} \nabla \cdot \boldsymbol{q}_b + \frac{1}{\rho_m c_p} \left(Q + \tau \frac{\partial Q}{\partial t} \right).$$

$$\begin{aligned} D_j &= D_j(T), \\ R_j &= R_j(T, E, C_1, C_2, \dots), \\ J_n, J_p &\text{ later depend on } T, \phi, \text{ and defect-assisted recombination.} \end{aligned}$$

Figure idea. A 2D silicon rectangle with localized internal heating, boundary-memory arrows, and coupling arrows to Module 3 and Module 5.

Physics drawing idea. Show a radiation-generated hot spot inside the domain, outward ballistic carrier paths, scattered thermal spreading, and side annotations: "updates defect kinetics" and "updates carrier transport coefficients."

Speaker emphasis. This is the project-specific landing slide. Explain clearly that Module 4 is now the thermal bridge between the defect-evolution physics and the later electrical-performance physics.

21 Slide 18: Validation strategy and the plots that should matter

Main question. How should Module 4 be validated before it is trusted inside the larger multiphysics chain?

Core message. Validation should start with simple canonical tests such as uniform equilibrium, localized pulse spreading, and boundary heating, while also keeping the original thin-film logic in

mind as a derivational benchmark. The important plots are the ones that distinguish ballistic-diffusive physics from ordinary local conduction.

Equations to show:

$$\frac{\partial^2 \theta_m}{\partial t^{*2}} + \frac{\partial \theta_m}{\partial t^*} = \frac{\mathrm{Kn}^2}{3}\frac{\partial^2 \theta_m}{\partial \eta^2} - \mathrm{Kn}\frac{\partial q_b^*}{\partial \eta}.$$

Figure idea. A validation slide with four boxes: thin-film benchmark logic, 2D uniform equilibrium, 2D localized pulse, and 2D boundary heating; beside them, expected output plots such as total temperature, medium temperature, ballistic flux, and comparison against Fourier or Cattaneo baselines.

Physics drawing idea. For the thin film, draw finite-arrival directions reaching an interior point. For the 2D case, show a hot region whose spread is shaped by both ballistic memory and scattering.

Speaker emphasis. Tell the audience exactly what would count as success: equilibrium stays uniform, localized pulse spreading is finite and physically sensible, boundary heating propagates inward with retained memory, and the model trends toward Fourier behavior in the strong-scattering limit.

22 Suggested presentation compression

If the final talk needs to be shorter, a good compressed sequence is:

1. Slide 1: project overview,
2. one selected module slide from Part I that best matches the audience,
3. Slide 9: why Fourier is not enough,
4. Slide 11: BTE starting point,
5. Slide 13 or 16: ballistic-diffusive split and final reduced PDE,
6. Slide 17: revised 2D Module 4 in `RadiationEffects_proj2`,
7. Slide 18: validation and expected plots.